Roll No: 5670432

KMU-MDCAT 25 CODE: D

Total number of MCQs: 180 Time: 3 hours No Negative Marking

BIOLOGY

1. Intrinsic factor secreted by parietal cells is essential for:
   a. Activation of pepsinogen to pepsin
   b. Absorption of vitamin B12 in the intestine
   c. Emulsification of dietary fats
   d. Neutralization of stomach acid in the duodenum
2. The Delta D cells in the Islets of Langerhans secrete:
   a. Insulin
   b. Somatostatin
   c. Glucagon
   d. Pancreatic polypeptide
3. Identify the main function of duodenum:
   a. Bile storage
   b. Absorption of water
   c. Digestion of food
   d. Waste storage
4. Which stage of the swallowing process prevents food or liquid from being aspirated to the lungs?
   a. Oral
   b. Pharyngeal
   c. Oesophageal
   d. Peristalsis
5. Which type of nephron is responsible for the development of osmotic gradients in the renal medulla?
   a. Glomerular
   b. Cortical
   c. Juxtamedullary
   d. Medullary
6. Which of the following best describes how the distal convoluted tubule contributes to regulation of blood pH in the body?
   a. Selective reabsorption of Glucose
   b. Active secretion of hydrogen ions into the filtrate
   c. Active secretion of sodium from the glomerular filtrate
   d. Tubular reabsorption of potassium ions
7. The sodium potassium pumps in the distal convoluted tubule are activated by:
   a. Aldosterone
   b. Antidiuretic hormone
   c. Anti-natriuretic peptide
   d. Renin
8. If a person drinks excessive amount of water, how does the kidney respond to maintaining osmoregulation?
   a. Increase ADH release and water reabsorption
   b. Increase aldosterone release and reabsorb more sodium
   c. Decrease ADH release and increase water excretion
   d. Decrease renin secretion and retain more water
9. The primary function of glomerular capillaries is:
   a. Reabsorption of water and solutes from the renal tubules
   b. Secretion of waste products from the blood into the tubules
   c. Filtration of blood from glomerular filtrate
   d. Supply of nutrients and oxygen to the kidney tissue
10. In a heat stroke the hypothalamus will detect an increase in core body temperature, which of the following response is triggered?
    a. Vasoconstriction and shivering
    b. Vasodilation and sweating
    c. Increased metabolic rate
    d. Release of thyroxin
11. Which nitrogenous waste has the lowest solubility in water?
    a. Urea
    b. Ammonia
    c. Uric acid
    d. Nitrite
12. By nature, human excretory system is:
    a. Ammoniotelic
    b. Uricotelic
    c. Ureotelic
    d. Aminotelic
13. Hyperparathyroidism may lead to the formation of:
    a. Calcium phosphate stones
    b. Uric acid stones
    c. Cystine stones
    d. Struvite stones
14. In somatic cell hybridization, which two cells are fused to produce monoclonal antibodies?
    a. T-cells and nerve cells
    b. B-lymphocytes and spleen cells
    c. T-cells and liver cells
    d. B-lymphocytes and macrophages
15. In molecular diagnostics, the primary function of DNA/RNA probes is to:
    a. Replace malfunctioning genes in effected cells
    b. Bind to specific DNA or RNA sequences for identification
    c. Cut genetic material at precise locations
    d. Stimulate immune responses against disease-causing organisms
16. A set of uniform proteins generated by genetically identical immune cells, derived from a single parent cell, are:
    a. Therapeutic proteins
    b. Immunoglobulins
    c. Monoclonal antibodies
    d. Recombinant proteins
17. Which biotechnological product is used to replace an abnormal gene in a patient's cells?
    a. Antibiotic
    b. Vaccine
    c. Vector
    d. Interferon
18. In Watson and Crick's DNA model, which of the following pairs with Cytosine?
    a. Adenine
    b. Guanine
    c. Thymine

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d. Uracil

19. Fats are considered as a very efficient source of energy because they:
    a. Produce ATP directly and without requiring respiration
    b. Enter into glycolysis without any modification
    c. Are highly oxidized compounds
    d. Generate multiple acetyl groups that produce more ATP
20. The junction between two consecutive neurons where information is transmitted from one neuron to the next is called:
    a. Node of Ranvier
    b. Synapse
    c. Axon terminal
    d. Dendrite cleft
21. All of the following are the functions of Golgi bodies EXCEPT:
    a. Processing of protein
    b. Plasma membrane formation
    c. Cellular respiration
    d. Lysosome formation
22. Which ion is tenfold higher in concentration outside the membrane of neuron during resting potential?
    a. Potassium
    b. Sodium
    c. Calcium
    d. Hydrogen
23. All of the following are the modes of transmission of AIDS, EXCEPT:
    a. Transfusion of infected blood
    b. Sharing infected needles
    c. Shaking hands with infected person
    d. Sexual contact with infected person
24. Which organelle of the cell is involved in the detoxification of toxins and poisonous compounds?
    a. Lysosomes
    b. Smooth Endoplasmic Reticulum
    c. Ribosomes
    d. Mitochondria
25. The active site is important in enzyme action because:
    a. It binds to the substrate
    b. It maintains the pH of reaction
    c. It provides energy for reaction
    d. It changes the shape of the enzyme
26. The idea that acquired characters are inherited is part of:
    a. Darwinism
    b. Neo Darwinism
    c. Lamarckism
    d. Biogenesis
27. Which disease is caused by enveloped virus?
    a. Leaf curl disease of cotton
    b. Flu
    c. Polio
    d. Hepatitis A
28. The glycoproteins are commonly found in:
    a. Mitochondria
    b. Chloroplasts
    c. Ribosomes
    d. Plasma membrane
29. Which structure enables the exchange of material between nucleus and cytoplasm?
    a. Plasma membrane
    b. Nuclear pores
    c. Lysosomes
    d. Mitochondria
30. The high specific heat capacity of water is due to:
    a. Ionic bonding
    b. Hydrogen bonding
    c. Covalent bonding
    d. Hydrophilic bonding
31. The contracted region of a chromosome that attaches to spindle fiber is called:
    a. Telomere
    b. Chromatin
    c. Centromere
    d. Nucleosomes
32. Touching a sharp object stimulates pain receptors. This information is carried to the central nervous system by the?
    a. Motor neuron
    b. Sensory neuron
    c. Associative neuron
    d. Effector neuron
33. The number of chromosomes in a haploid cell are:
    a. Half the chromosomes in a normal body cell
    b. Double the chromosomes in a normal body cell
    c. Quarter the chromosomes in a normal body cell
    d. Equal to the chromosomes in a normal body cell
34. Which of the following is a branched polysaccharide found in animals:
    a. Cellulose
    b. Glycogen
    c. Amylose
    d. Chitin
35. A substance that binds to an enzyme, but NOT at the active site and reduces the enzyme activity is called a:
    a. Competitive inhibitor
    b. Substrate
    c. Non-Competitive inhibitor
    d. Cofactor
36. On the basis of morphological classification, influenza virus is an example of:
    a. Helical capsid virus
    b. Polyhedral capsid virus
    c. Enveloped capsid virus
    d. Non-enveloped capsid virus
37. Which of the following is NOT a globular protein?
    a. Enzyme
    b. Albumen
    c. Hemoglobin
    d. Collagen
38. The state, when a neuron is NOT conducting an impulse during resting membrane potential is called;
    a. Polarized
    b. Depolarized
    c. Repolarized
    d. Hyperpolarized

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39. Which of the following waves travel along a neuron during nerve impulse conduction?
a. Thermal waves
b. Magnetic waves
c. Electromagnetic waves
d. Electrochemical waves

40. The reticular formation in the brain runs through which specific regions?
a. Forebrain and Midbrain
b. Hindbrain and Midbrain
c. Cerebellum and Forebrain
d. Telencephalon and Cerebellum

41. Enzymes increase the rate of reaction by:
a. Increasing activation energy
b. Lowering activation energy
c. Increasing pH
d. Decreasing pH

42. A diabetic patient is advised to avoid both sucrose and lactose, because they both:
a. Are structural carbohydrates
b. Increase blood glucose after hydrolysis
c. Cannot be digested in human
d. Act as non-caloric sweeteners

43. Match the CORRECT structure of the brain with its function:
a. Medulla: Breathing
b. Pons: Memory
c. Cerebellum: Dreaming
d. Midbrain: Balance

44. Living cells Do NOT directly acquire energy released from the breakdown of food molecules because:
a. Glucose cannot be broken down inside the cells
b. Energy released is too small to be used by the cells
c. Glucose molecules do not store any energy
d. The energy released is too large, leading to heating and wastage

45. Which properties of water enable it to circulate in living bodies and act as transport medium?
a. Ionization and low density
b. Cohesion and Ionization
c. Ionization and adhesion
d. Adhesion and cohesion

46. A scientist cuts the tails of mice for several generations, but their offspring are always born with tails. This rejects:
a. Darwinism
b. Neo Darwinism
c. Lamarckism
d. Mutation theory

47. How are the phospholipid molecules arranged in the plasma membrane?
a. Hydrophilic heads face inwards and hydrophobic tails face outwards
b. Both Hydrophilic heads face each other in the membrane
c. Hydrophilic heads face outwards and hydrophobic tails face inwards
d. Hydrophilic heads and hydrophobic tails are randomly distributed

48. The structure of RNA consists of:
a. Double polynucleotide strand
b. Deoxyribose sugar
c. Five different types of nucleotides
d. Base uracil instead of thymine

49. The constant diameter of DNA is maintained by pairing:
a. Purine towards Pyrimidine
b. Pyrimidine towards Pyrimidine
c. Purine towards Purine
d. Sugar towards Phosphate

50. Identify the CORRECT option that matches the sugar with its carbon number and functional group:
a. Glyceraldehyde - Triose, Ketone group
b. Ribose - Pentose, Aldehyde group
c. Ribulose - Pentose, Aldehyde Group
d. Galactose - Hexose, Ketone Group

51. The CORRECT sequence of events in a reflex arc is:
a. Receptor, associative neuron, sensory neuron, motor neuron, effector
b. Receptor, motor neuron, associative neuron, sensory neuron, effector
c. Receptor, sensory neuron, associative neuron, motor neuron, effector
d. Receptor, sensory neuron, motor neuron, associative neuron, effector

52. Fertilisation normally occurs in the:
a. Uterus
b. Ovary
c. Vagina
d. Fallopian tubes

53. Which cells of the fallopian tube moisten and nourish the ovum?
a. Ciliated epithelial cells
b. Non ciliated epithelial cells
c. Germ cells
d. Endometrial cells

54. Which of the following is NOT the accessory gland of human male reproductive system?
a. Bulbourethral gland
b. Seminal vesicle
c. Prostate gland
d. Testes

55. Syphilis is caused by what kind of bacteria:
a. Spirochaete
b. Cocci
c. Bacillus
d. Vibrio

56. What is FALSE about cartilages?
a. Cells are called chondrocytes
b. Consist of Type II collagen
c. Heal very slowly
d. Have an extensive blood supply

57. Bones provide a rigid framework with an inorganic matrix of:
a. 35%
b. 45%
c. 55%
d. 65%

58. An important feature of bone remodeling is bone breakdown. Which cell carries out this function?
a. Chondrocyte
b. Osteocyte

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c. Osteoclast
d. Osteoblast

59. Myofibrils consist of small contractile units called___
a. Sarcoplasm
b. Sarcolemma
c. Sarcomere
d. Sarcoplasmic reticulum

60. Which muscle type is under conscious control and is multinucleated?
a. Smooth muscle
b. Skeletal muscle
c. Cardiac muscle
d. Ciliary muscle

61. Which part of the sarcomere contain both actin and myosin filament?
a. I-band
b. Z-line
c. A-band
d. H-zone

62. The primary role of calcium ions in muscle contraction is to:
a. Produce energy
b. Bind with troponin
c. Carry nerve impulses
d. Breakdown ATP

63. Which of the following is an example of a fibrous joint?
a. Shoulder joint
b. Elbow joint
c. Intervertebral discs
d. Skull sutures

64. Which of Mendel's laws can best explain why a child may inherit brown eyes even if one parent has blue eyes?
a. Law of Dominance
b. Law of Segregation
c. Law of Independent Assortment
d. Law of Recombination

65. What is the significance of the 9:3:3:1ratio in a dihybrid cross?
a. It proves that all genes are linked
b. It demonstrates that traits assort independently
c. It indicates co-dominance between alleles
d. It confirms that mutations have occurred

66. In crossing over, an exchange of maternal and paternal chromatid parts occurs while homologous chromosomes are paired during _____stage of meiosis.
a. Metaphase I
b. Prophase I
c. Anaphase II
d. Telophase I

67. Linked genes DO NOT follow Mendel's Law of Independent Assortment because___.
a. They are located on different chromosomes
b. They always undergo crossing over
c. They are physically close together on the same chromosome
d. They rarely separate during meiosis

68. A carrier female for an X-linked recessive disorder:
a. Expresses the disorder fully
b. Cannot pass the disorder to offspring
c. Can pass the disorder to her sons
d. Only passes the disorder to daughters

69. Which of the following is an X-linked recessive disorder in humans?
a. Cystic fibrosis
b. Thalassemia
c. Hemophilia
d. Sickle cell anemia

70. Which of the following CORRECTLY describes the expected outcome of children from a carrier mother for haemophilia (X^HX^h) and a normal father (X^HY)?
a. All sons will have hemophilia
b. All daughters will be carriers
c. 50% of sons will have hemophilia, and 50% of daughters will be carriers
d. All offspring will be unaffected

71. The heart is surrounded by a tough, inelastic double membranous covering called:
a. Pleura
b. Peritoneum
c. Pericardium
d. Meninges

72. Which phase of the cardiac cycle is characterized by the opening of semilunar valves?
a. Atrial systole
b. Atrial diastole
c. Ventricular systole
d. Ventricular diastole

73. Which part of the heart's conducting system delays the impulse from atria to the ventricles?
a. Sino-atrial node
b. Atrio-ventricular node
c. Purkinje fibers
d. Atrio-ventricular valves

74. The heartbeat sound "LUB" is produced on closure of:
a. Aortic valve
b. Atrio-ventricular valves
c. Pulmonary valves
d. Semilunar valves

75. What is the primary function of lymphatic vessels?
a. Transport oxygenated blood
b. Drain excess interstitial fluid
c. Produce antibodies
d. Store metabolic waste

76. The primary role of helper T-cells in immune response is:
a. Secreting perforins to destroy target cells
b. Assisting B-cells and other T-cells in their function
c. Inhibiting over activity of the immune system
d. Engulfing and digesting pathogens

77. Identify the primary characteristic of 3rd line defense.
a. Complement system and interferons
b. Physical component of skin defense
c. Specific recognition and memory of pathogens
d. General inflammation at the site of infection

78. Which of the following best describes Inspiratory Reserve Volume (IRV)?

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a. The volume of air forcibly exhaled after normal tidal volume
b. The volume of air remaining in the lungs after forcible expiration
c. The volume of air exchanged during normal breathing
d. The volume of air forcibly inhaled after normal tidal volume

79. During gaseous exchange in human respiration, enzyme carbonic anhydrase is directly involved in the:
a. Release of oxygen from hemoglobin
b. Combination of water with carbon dioxide
c. Breakdown of oxyhemoglobin
d. Binding of oxygen to myoglobin

80. What happens when the external intercostal muscles contract during inhalation in humans?
a. The rib cage compresses
b. The ribs move outward
c. The sternum moves inwards
d. The diaphragm muscles relax

81. The primary effect of smoking on the respiratory system:
a. Increased lung function due to presence of nicotine
b. Reduced risk of lung cancer due to tolerance
c. Damage to alveoli reducing gaseous exchange
d. Improves oxygenation of the blood by activating hemoglobin

PHYSICS

82. According to Lenz's Law, the direction of induced current in a conductor is such that it:
a. Opposes the change in magnetic flux
b. Enhances the change in magnetic flux
c. Is perpendicular to the magnetic field
d. Is parallel to the magnetic field

83. In a pure capacitance AC circuit, the current:
a. Lags behind voltage by 90°
b. Leads the voltage by 90°
c. Is in phase with the voltage
d. Leads the voltage by 45°

84. In a full-wave rectifier using two diodes, the diodes D_1 and D_2 operate:
a. Simultaneously during both half-cycles
b. In alternate switching mode
c. Only when both ends of transformer are positive
d. Only in reverse bias condition

85. The energy (E) of a quantum is given by which equation?
a. $E = mc^2$
b. $E = hf$
c. $E = \frac{1}{2}mv^2$
d. $E = qV$

86. In hydrogen spectrum, the Brackett series lies in
a. Ultraviolet region
b. Visible region
c. Infrared region
d. X ray region

87. Which of the following statement best describes the nature of nuclear decay?
a. It occurs both spontaneously and randomly
b. It occurs only when external energy is supplied
c. It occurs at regular time intervals for each atom
d. It can be controlled by temperature and pressure

88. A boat moves 4 km east and then 3 km north. Another boat moves 3 km north first and then 4 km east. Which statement is CORRECT about their final displacements?
a. Both boats have the same displacement vector
b. The first boat's displacement is greater because it traveled east first
c. The second boat's displacement is greater because it travelled north first
d. Both boats end at different positions but cover the same distance

89. The area under the line on a displacement-time graph of a car moving with uniform velocity would be:
a. Rectangle
b. Triangle
c. Trapezium
d. Parallelogram

90. A car starts from rest and moves with a uniform acceleration of $3m/s^2$. What will be its velocity after 5 seconds?
a. 8m/s
b. 12m/s
c. 15m/s
d. 18m/s

91. A passenger is standing in a stationary bus. When the bus suddenly starts moving forward, the passenger falls backward. Which phenomenon best explains this observation?
a. Friction
b. Gravity
c. Inertia
d. Deceleration

92. A ball is projected at an angle 45° with an initial speed of 20 m/s on earth. How does R_W (Range without air resistance) compare to R_A (Range with resistance).
a. R_A will be greater than R_W because air resistance reduces horizontal speed
b. R_W will be equal to R_A because gravity is unchan
c. R_A will be lesser than R_W because air resistance reduces horizontal speed
d. R_W will be lesser than R_A because air resistance random

93. A 0.02 kg bullet moving at 300m/s embeds itself i a 2 kg block at rest on a smooth surface. What is velocity of the block-bullet system just after an impact?
a. 2m/s
b. 3m/s
c. 4m/s
d. 5m/s

94. Which of the following statement about projectil motion is CORRECT?

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a. The horizontal velocity of a projectile changes constantly due to gravity
b. The vertical velocity of a projectile remains constant throughout the flight
c. At the highest point, the vertical velocity of the projectile is zero, but the horizontal velocity remains unchanged
d. The acceleration of the projectile is zero at the peak of its trajectory

95. The rate of doing work at any instant of time is called:
a. Work done
b. Instantaneous power
c. Average power
d. Mechanical energy

96. A 5kg body falls from the height of 30m towards the ground. All its potential energy is converted into heat on impact. What is the heat energy produced?
a. 1270J
b. 1370J
c. 1470J
d. 1570J

97. Two students, A and B, each carry a 20kg load to the top of a 10m high staircase. Student A takes 10sec, while student B takes 20sec. Which statement is CORRECT?
a. Student A does more work than student B
b. Student B uses more power than student A
c. Both students do the same amount of work, but student A uses more power
d. Student A and B use the same power since they lifted the same weight

98. A wheel of radius 0.4m has an angular acceleration of 6 rad/s². The linear acceleration is:
a. 1.2 m/s²
b. 1.2 m/s
c. 2.4 m/s
d. 2.4 m/s²

99. If an object is moving anticlockwise along a circular path, in a horizontal plane on a page then the direction of its angular velocity is:
a. Tangential to any point on the circle
b. Towards centre of the circle
c. Perpendicular to the plane and pointing out of the page
d. Perpendicular to the plane and pointing into the page

100. If a particle moves along a circular path of radius r with angular displacement θ (in radians), then the arc length s is given by:
a. r×θ
b. θ/r
c. r²×θ
d. θ/r²

101. A ball of weight F_G is falling vertically downward through air. If the drag force acting on it at some instant is F_D, what is the F_{net} (net force) on the ball?
a. F_G+F_D
b. F_G-F_D
c. F_D-F_G
d. F_D/F_G

102. Most kinds of fluid flow are turbulent rather than laminar because of:
a. Zero viscosity
b. Very low velocities
c. High velocities
d. No resistance

103. According to the equation of continuity, when the cross-sectional area of a pipe decreases, the fluid velocity:
a. Increases
b. Decreases
c. Remains the same
d. Becomes zero

104. Water flows steadily through a pipe that gradually narrows. At the wider end, the velocity of water is 1m/s. At the narrower end, the velocity is 3m/s. Which statement is CORRECT about the pressure in the narrower end compared to the wider end?
a. Pressure is lower at the narrow end because velocity is higher
b. Pressure is higher at the narrow end because velocity is higher
c. Pressure is the same at both ends since flow is continuous
d. Pressure is independent of velocity of water

105. What is the necessary condition of a wave motion?
a. The medium must be elastic
b. The medium must be inelastic
c. The particles of the medium must be independent of each other
d. The particles of the medium must not be dependent on each other

106. A progressive wave is one which:
a. Does not vibrate the medium
b. Carries energy across the medium
c. Propagates only through air
d. Requires a denser medium for propagation

107. Speed of sound in air increases with:
a. Higher temperature, higher humidity
b. Lower temperature, lower humidity
c. Higher pressure at constant temperature
d. Higher density at constant elasticity

108. A wave has velocity 300m/s and frequency 100Hz. If the medium is changed so that velocity doubles but frequency remains constant, the new wavelength will be:
a. Halved
b. Doubled
c. Same
d. Zero

109. When a particle executing simple harmonic motion moves from the mean position to the extreme position, its kinetic energy:
a. Increases continuously
b. Decreases continuously and becomes zero at the extreme position
c. Remains constant throughout the motion
d. Becomes maximum at the extreme position

110. Heat will spontaneously flow from:
a. Lower to higher internal energy only
b. High pressure to low pressure

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c. Cold to hot
d. Hot to cold

111. The SI unit of molar specific heat is:
a. J mole⁻¹K⁻¹
b. J mole K⁻¹
c. J mole K⁻¹
d. J mole⁻¹K

112. At a constant Pressure (C_p) and Volume (C_v), an ideal gas has C_v=3/2R. Then C_p will be?
a. R
b. 3/2R
c. 5/2R
d. 7/2R

113. The electric field at a point is defined as:
a. Potential per unit charge
b. Work done per unit time
c. Charge per unit area
d. Force per unit positive test charge

114. Coulomb's law fits well into:
a. Newton's 1st law
b. Newton's 2nd law
c. Newton's 3rd law
d. Gauss's Law

115. The work done in moving a unit positive charge from one point to another while keeping the charge in electrostatic equilibrium is called:
a. Kinetic energy
b. Potential energy
c. Elastic potential energy
d. Potential difference

116. If the potential difference (V) across a conductor is doubled, keeping resistance (R) constant, the power dissipated(P) becomes:
a. Doubled
b. Halved
c. Four times
d. Remains unchanged

117. When the area vector A is parallel to the magnetic field B, what is the value of magnetic flux (Φ_B)?
a. 0
b. BA cos 0°
c. BA cos 90°
d. B/A

CHEMISTRY

118. Consider the chlorination of methane to methyl chloride, the attack of chlorine free radical on methane, occurs in which phase?
a. Before initiation
b. Initiation
c. Propagation
d. Termination

119. A patient with pancreatic insufficiency shows reduced activity of an enzyme that hydrolyzes the first peptide bond at the carboxyl end of proteins and peptides. Which enzyme is deficient?
a. Elastase
b. Pepsin
c. Carboxypeptidase
d. Collagenase

120. The force of attraction due to temporary dipoles is:
a. Dipole-dipole
b. Dispersion force
c. Debye force
d. Ion-dipole force

121. In which case does a chemical reaction proceed nearly to completion?
a. When Kc is small but positive
b. When Kc is large but positive
c. When Kc is negative
d. When Kc is approximately equal to 1

122. A student adds bromine water to ethene and observes rapid decolourisation. What makes ethene more reactive than ethane in this reaction?
a. Ethene has a higher molecular mass
b. Ethene contains a weak and exposed pi bond
c. Ethane contains fewer sigma bonds
d. Ethene undergoes substitution more readily

123. What best describes the overall reaction in benzene?
a. Addition of an electrophile across a double bond
b. Substitution of a halogen by an nucleophile
c. Substitution of a proton (H^+) by an electrophile on the aromatic ring
d. Substitution of a methyl group by a nucleophile

124. Real gases DO NOT reach absolute zero in practice because:
a. Molecular collisions become inelastic due to increased kinetic energy
b. Intermolecular forces become negligible and molecules disperse
c. Kinetic energy of molecules increases due to compression
d. Intermolecular forces exceed kinetic energy of molecules

125. The energy of an orbital is determined by:
a. Hund's Rule
b. Pauli exclusion principle
c. n+l rule
d. Boyles principle

126. The Ksp value of salt $AB \rightleftharpoons A^+ + B^-$ is 9×10^{-6}. Its molar solubility will be:
a. 3×10^{-6}
b. 9×10^{-3}
c. 3×10^{-3}
d. 9×10^{-6}

127. Water changes from a liquid at 4°C to ice at 0°C. What is the change in volume?
a. 9% increase
b. 9% decrease
c. 19% increase
d. 19% decrease

128. For an exothermic reaction, the energy level of reactant is:
a. Less than the product
b. More than the product
c. Equal to the product
d. Zero

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129. In an electrolytic cell when the current passes through a solution, the anode is:
   a. A positive electrode where oxidation occurs
   b. A negative electrode where oxidation occurs
   c. A positive electrode where reduction occurs
   d. A negative electrode where reduction occurs

130. An increase in the internal energy of a chemical system can lead to all EXCEPT:
   a. An increase in temperature due to rise in kinetic energy of molecules
   b. A phase change such as melting or evaporation
   c. A chemical reaction if energy supplied is sufficient to break bonds
   d. An increase in temperature due to drop in kinetic energy of molecules

131. In ethene molecule, each carbon atom has three hybridized sp^2 orbitals which are:
   a. Coplanar
   b. Tetrahedral
   c. Linear
   d. Pyramidal

132. Which of the following best explains the reaction between Beryllium (Be) and Oxygen (O)?
   a. Be burns vigorously with oxygen forming a layer of BeO, which accelerates the oxidation of remaining metal
   b. Be reacts with oxygen forming a layer of BeO, which protects the metal from further oxidation
   c. Be reacts slowly with oxygen to form a volatile oxide BeO, which evaporates quickly
   d. Be is the only alkaline earth metal that doesn't react with oxygen

133. The value of R in $atm.dm^3.mol^{-1}.K^{-1}$ is:
   a. 0.0821
   b. 0.821
   c. 62.4
   d. 8.314

134. The rate of a chemical reaction changes with:
   a. Concentration of reactant molecules
   b. Concentration of product molecules
   c. Concentration of both reactant and product
   d. Rate constant

135. According to Planck's quantum theory, if the frequency of photon is doubled, the value of "h" will be?
   a. Doubled
   b. Increased 3 times
   c. Increased 4 times
   d. Unchanged

136. The specific rate constant (k) of a reaction is related to the concentration of reactants:
   a. Directly
   b. Inversely
   c. Exponentially
   d. Independently

137. An experiment shows that heating a 'protein' disrupts the alpha helix structure. Which protein structure is mainly affected?
   a. Primary
   b. Secondary
   c. Tertiary
   d. Quaternary

138. Which of the following is NOT a postulate of kinetic molecular theory of gases?
   a. Gas molecules undergo elastic collision
   b. Gas molecules are in continuous random motion
   c. Gas molecules do not exert pressure when molecules collide with wall of container
   d. Gas molecules are far away from each other

139. Spectral series for hydrogen spectrum are :
   a. 2
   b. 3
   c. 5
   d. 7

140. The most electronegative element in periodic table is:
   a. F
   b. Cl
   c. O
   d. N

141. If ΔH for a reaction is positive, then by decreasing temperature, the reaction will :
   a. Move forward
   b. Move in reverse
   c. Have no effect
   d. Be both forward and reverse

142. The addition of water to propene in the presence of sulfuric acid produces:
   a. Propanol
   b. Propan-2-ol
   c. Butanol
   d. Ethanol

143. In which of the following molecules does the central atom use sp^2 hybridization?
   a. PH_3
   b. NH_3
   c. CH_3^+
   d. SbH_3

144. In a molecule of phenol, the carbon atom which is attached to -OH group is:
   a. sp hybridized
   b. sp^2 hybridized
   c. sp^3 hybridized
   d. unhybridized

145. Which one of the following is a planar molecule?
   a. NH_3
   b. H_2O
   c. BF_3
   d. CH_4

146. Which of the following has highest bond energy?
   a. HCl
   b. HI
   c. HF
   d. HBr

147. Identify the CORRECT electronic configuration for an element with atomic number 24?

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a. $1s^1 2s^1 2p^4 3s^1 3p^4 3d^4$
b. $1s^1 2s^1 2p^4 3s^1 3p^4 4s^1 3d^4$
c. $1s^1 2s^1 2p^4 3s^1 3p^4 4s^1 3d^5$
d. $1s^1 2s^1 3s^1 2p^4 4s^1 3p^4 3d^5$

148. Balance the equation, using the oxidation number method for: $Cu + H_2SO_4 \rightarrow CuSO_4 + SO_2 + H_2O$
a. $Cu + H_2SO_4 \rightarrow CuSO_4 + SO_2 + H_2O$
b. $Cu + H_2SO_4 \rightarrow CuSO_4 + SO_2 + H_2$
c. $Cu + 2H_2SO_4 \rightarrow CuSO_4 + SO_2 + 2H_2O$
d. $2Cu + 2H_2SO_4 \rightarrow 2CuSO_4 + SO_2 + 2H_2O$

149. The compound that reacts with phenyl hydrazine to form a crystalline phenyl hydrazone derivative is:
a. Butanal
b. 1,3-butadiene
c. Ethyl acetate
d. Ethanol

150. If 4g of H_2 reacts with 2 moles of O_2 to form water, which one is the excess reagent?
a. H_2 only
b. O_2 only
c. H_2O only
d. Both O_2 and H_2

151. If 10 moles of magnesium react with excess of oxygen. Calculate the theoretical yield of magnesium oxide (MgO). (Molar mass of Magnesium=24g/mol, Oxygen=16g/mol)
a. 160g
b. 240g
c. 320g
d. 400g

152. Acetaldehyde reacts with ethanol in presence of acid and initially produces:
a. Acetal
b. Hemiacetal
c. Diol
d. Diethoxyethane

153. IUPAC name of given compound $Br-CH_2-CH_2-CO-CH_3$ is?
a. 1-bromobutan-1-one
b. 1-bromobutan-2-one
c. 1-bromobutan-3-one
d. 1-bromobutan-4-one

154. Consider the given reaction: $2Na + Cl_2 \rightarrow 2NaCl$
If 4 moles of Na and 2 moles of Cl_2 are reacted, how much Cl_2 will remain unreacted?
a. 0 mol
b. 0.5 mol
c. 1 mol
d. 1.5 mol

155. The electronic configuration of Fe^{3+} (Z=26) is:
a. $[Ar]4s^2, 3d^3$
b. $[Ar]4s^1, 3d^4$
c. $[Ar]4s^0, 3d^5$
d. $[Ar]4s^0, 3d^6$

156. If pressure and temperature of a gas are doubled, the new volume will be:
a. Doubled
b. Halved
c. Tripled
d. Same

157. IUPAC name of $CH_3CH_2C\equiv CH$ is:
a. hex-1-yne
b. pent-2-yne
c. pent-4-yne
d. pent-1-yne

158. According to Le Chatelier's Principle, when the pressure of a gaseous equilibrium system is increased, the equilibrium shifts towards:
a. No change in equilibrium position
b. The side with greater volume
c. The side with lower volume
d. The side with more moles of gas

159. Consider a reaction $2A + B_2 \rightarrow 2AB$. Which of the following mixtures would make A the limiting reagent?
a. 300 atoms of A and 400 molecules of B_2
b. 100 atoms of A and 50 molecules of B_2
c. 2 mol of A and 1 mol of B_2
d. 5 mol of A and 2.5 mol of B_2

160. On increasing the temperature, the rate of reaction increases mainly because:
a. The activation energy of the reaction increases
b. The concentration of the reacting molecules increases
c. The collisions frequency increases
d. The energy of molecules decreases

161. Heat of sublimation of Na= 108kJ/mol, Ionization energy of Na = 496 kJ/mol, Bond dissociation energy of Cl_2 = 121 kJ/mol, Electron affinity of Cl= -349 kJ/mol and Enthalpy of formation of NaCl= -411 kJ/mol. Calculate the lattice energy of NaCl:
a. -678 kJ/mol
b. -727 kJ/mol
c. -819 kJ/mol
d. -832 kJ/mol

162. The molecular orbitals in benzene are:
a. Localized
b. Delocalized
c. Hybridized
d. Polarized

ENGLISH

163. In the sentence "Jogging every morning improves my mood," what role does the gerund phrase play?
a. Object of a verb
b. Direct object
c. Predicate noun
d. Subject

164. The meeting has been scheduled ___ 3 p.m. sharp.
a. on
b. at
c. in
d. to

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165. "She studied hard; _____, she passed the exam with distinction." Choose the CORRECT transitional device:
   a. however
   b. for instance
   c. consequently
   d. in contrast

166. She completed the task with great difficulty. Identify the CORRECT usage of passive voice.
   a. The task was completed with great difficulty by her.
   b. The task has been completed with great difficulty by her.
   c. The task was been completed with great difficulty by her.
   d. The task was being completed with great difficulty by her.

167. Each of the boys _____ ambitious to lead the team.
   a. have
   b. has
   c. is
   d. are

168. "Revenge is a kind of wild justice." The sentence illustrates the use of:
   a. Personification
   b. Pun
   c. Metaphor
   d. Hyperbole

169. The old man was feeble, barely able to walk. The word "feeble" means:
   a. Healthy
   b. Weak
   c. Strong
   d. Fat

170. Which literary device involves using words or phrases that convey the opposite of their literal meaning?
   a. Irony
   b. Parody
   c. Satire
   d. Sarcasm

171. All of the players forgot _____ jerseys.
   a. his
   b. her
   c. there
   d. their

LOGICAL REASONING

172. A medication dose starts at 100mg and halves each day. What will be the dose on day 4?
   a. 12.5mg
   b. 25mg
   c. 30mg
   d. 50mg

173. A library has 4 books on Shelf A, and 8 books on Shelf B. Some books are removed from Shelf B such that both shelves now have exactly the same number of books, How many books were removed from Shelf B?
   a. 2 books
   b. 3 books
   c. 4 books
   d. 6 books

174. Ahmed ranks 10th in the class of 46 students. There are only 7 students below Bilal rank wise. How many students are there between Ahmed and Bilal?
   a. 27
   b. 28
   c. 30
   d. 32

175. How is Ali related to Mustafa? If Ali says, "Mustafa's mother is the only child of my grandmother."
   a. Brother
   b. Cousin
   c. Uncle
   d. Father

176. A mobile company is deciding whether to collect personal user data for improving advertisement. Which of the following is a moral argument against collecting the data?
   a. Users may lose trust in the company, which could hurt profits.
   b. It is wrong to collect data without users' clear and informed consent.
   c. Competitors already collect more data, and we need to stay competitive.
   d. More data means more accurate advertising, which boosts sales.

177. A father is 5 years older than twice his son's age. If the son is 12, what is the father's age?
   a. 28 years
   b. 29 years
   c. 30 years
   d. 32 years

178. Which number comes in the missing place? 200, 180, 162, 146, 132, _____.
   a. 118
   b. 120
   c. 122
   d. 126

179. If 5 boxes of soap weigh 75 kilos and each box when empty weighs 3 kilos. What is total weight of the soaps?
   a. 15
   b. 30
   c. 45
   d. 60

180. Consider the sequence: 64, 32, 16, 8, . . . What is the 10th term of this sequence?
   a. 2
   b. 4
   c. 1/4
   d. 1/8

Good Luck! The END.

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KMU-MDCAT 2025 KEYS

	KEY A		KEY B		KEY C		KEY D
1	d	1	b	1	c	1	b
2	b	2	d	2	b	2	b
3	d	3	a	3	b	3	c
4	a	4	b	4	b	4	b
5	d	5	b	5	b	5	c
6	d	6	c	6	c	6	b
7	c	7	c	7	d	7	a
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22	b	22	b	22	c	22	b
23	b	23	c	23	a	23	c
24	b	24	b	24	b	24	b
25	b	25	c	25	b	25	a
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31	b	31	c	31	c	31	c
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33	c	33	a	33	c	33	a
34	b	34	b	34	b	34	b
35	c	35	b	35	a	35	c
36	b	36	c	36	c	36	c
37	a	37	c	37	b	37	d
38	c	38	b	38	d	38	a

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	KEY A		KEY B		KEY C		KEY D
39	c	39	d	39	b	39	d
40	b	40	b	40	b	40	b
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44	b	44	b	44	b	44	d
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46	c	46	c	46	c	46	c
47	c	47	b	47	d	47	c
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49	d	49	b	49	d	49	a
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60	b	60	b	60	b	60	b
61	c	61	b	61	c	61	c
62	b	62	b	62	d	62	b
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70	b	70	b	70	b	70	c
71	b	71	c	71	c	71	c
72	b	72	d	72	b	72	c
73	a	73	b	73	d	73	b
74	d	74	d	74	a	74	b
75	d	75	a	75	b	75	b
76	c	76	d	76	b	76	b

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	KEY A		KEY B		KEY C		KEY D
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79	a	79	c	79	c	79	b
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81	c	81	c	81	c	81	c
82	a	82	d	82	b	82	a
83	b	83	c	83	b	83	b
84	c	84	a	84	d	84	b
85	c	85	b	85	a	85	b
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91	c	91	a	91	b	91	c
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93	c	93	b	93	b	93	b
94	a	94	d	94	b	94	c
95	b	95	a	95	b	95	b
96	c	96	c	96	c	96	c
97	a	97	d	97	a	97	c
98	a	98	c	98	a	98	d
99	a	99	d	99	b	99	c
100	b	100	c	100	c	100	a
101	a	101	b	101	c	101	b
102	b	102	a	102	c	102	c
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104	d	104	b	104	c	104	a
105	a	105	b	105	b	105	a
106	c	106	c	106	c	106	b
107	d	107	a	107	c	107	a
108	c	108	a	108	d	108	b
109	d	109	b	109	c	109	b
110	c	110	c	110	a	110	d
111	b	111	c	111	b	111	a
112	a	112	c	112	c	112	c
113	b	113	b	113	a	113	d
114	b	114	c	114	a	114	c

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	KEY A		KEY B		KEY C		KEY D
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127	b	127	b	127	b	127	a
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148	c	148	b	148	c	148	c
149	c	149	a	149	c	149	a
150	b	150	d	150	a	150	b
151	b	151	a	151	b	151	d
152	b	152	b	152	b	152	b

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	KEY A		KEY B		KEY C		KEY D
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178	b	178	b	178	d	178	b
179	a	179	b	179	a	179	d
180	b	180	d	180	c	180	d

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